Non-violent emergency call outcome analysis in Eugene, Oregon

Investigating Eugene's response to addiction, homelessness and mental health related crisis

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Contents

1 Introduction

In this document, we will be conducting a quantitative analysis of the call outcomes for mental health, addiction and homelessness related emergencies in Eugene. We will be focusing on the code and methods. For more details about the project, please find the full report at the root of the repository: `internship_report.pdf`.

2 Work Environment

2.1 Prerequisites

- R (version 4.5 or higher recommended)
- RStudio (optional, but recommended)

2.2 Managing Dependencies

This project uses `renv` for R package management. To install all required packages:

1. Open the project in RStudio or start an R session in the project directory.
2. Run: `R renv::restore()` This will install all packages listed in the project's lockfile.

If you add new packages, use `renv::snapshot()` to update the lockfile.

3 Data

The data used in this analysis is processed and merged from the following sources. The final dataset includes every agency (EPD, SPD, CAHOOTS and MCSLC), and dispatched calls only, from 2015-01-01 to 2025-12-31.

3.1 Available raw data

- SPD Call for Services, 2015-2025 : .xlsx document, one sheet per year
- SPD Responding Units, 2015-2025 : .xlsx document, one sheet per year
- SPD Call Outcome, 2015-2025 : .xlsx doxument, key codes and one sheet per year
- MCS LC data, 2024/08/18 - 2025/12/17 : .xlsx document, one sheet
- Eugene CAD data, 2015-2025 : one .csv per year

3.1.1 SPD data

This dataset contains detailed records of calls for service for Springfield Police Department from 2015 to 2025, including temporal management of calls, the nature of the interventions, and the mobile resources deployed.

There are 14 columns and 567,655 rows.

Variable	Type	Example Value	Description
timestamp	datetime	2015-01-01 01:16:21	Date and time when the incident was initially recorded.
incident_id	double	15000074	Unique identifier for each incident.
agency	factor	SPD	Agency responsible for the intervention.
city	factor	Springfield	City.
nature_code	factor	ALMAUD	Coded category for the type of call received.
nature_final	factor	AUDIBLE ALARM	Final classification of the incident after assessment.
close_code	factor	BLDS	Coded reason or outcome for closing the incident.
close_code_definition	factor	Building Check Secure	Full description of the incident closure status.
priority	factor	3	Numerical priority level assigned to the call.
prime_unit	character	1S18	Primary unit assigned to respond to the incident.
units	character	1S18, 1S21, K94	List of all units dispatched to the scene.
dispatch_time	time	01:18:00	Time at which units were dispatched.
arrival_time	time	01:21:00	Time at which the first unit arrived at the scene.
clear_time	time	01:32:00	Time at which the incident was marked as completed.
call_source	factor	PHONE	Origin or method of the incoming call.

3.1.2 Eugene CAD data

This dataset contains detailed records of calls for service for Eugene from 2015 to 2025, including temporal management of calls, the nature of the interventions, and the mobile resources deployed.

There are 17 columns and 1,446,014 rows.

Variable	Type	Example Value	Description
timestamp	datetime	2015-01-01 00:00:00	Date and time when the call to report an incident was initially recorded.
incident_id	double	15000001	Unique identifier for each incident.
agency	factor	EPD	Agency responsible for the intervention (e.g., Eugene Police Department).
service	factor	LAW	Type of service mobilized (e.g., Law Enforcement, Fire, Other).
call_source	factor	SELF	Origin of the call (e.g., Self-initiated, Phone, W911).
nature	factor	PERSON STOP	Nature or reported motive of the incident.
close_code	factor	ASST	Abbreviated closing code indicating the outcome of the intervention.
closed_as	factor	ASSISTED	Full description of how the incident was resolved.
priority	factor	6	Priority level assigned to the incident ranging from 1 (highest) to 9 (lowest).
dispatch_time	time	00:00:00	Time when units were dispatched to the incident location.
arrival_time	time	00:00:00	Time when the first unit arrived on the scene.
clear_time	time	00:03:37	Time when the incident was marked as cleared or completed.
dispatch	factor	1	Binary indicator specifying whether a unit was dispatched.
arrival	factor	1	Binary indicator specifying whether a unit arrived on the scene.
prime_unit	factor	_5E48	Identifier of the primary unit assigned to the incident.
nb_units_dispatched	double	1	Total number of units dispatched to the intervention.

Variable	Type	Example Value	Description
nb_units_arrived	double	1	Total number of units that successfully reached the scene.

3.1.3 Mobile Crisis Services - Lane County data

This dataset tracks crisis intervention activities from MCS LC, focusing on response timelines, and clinical outcomes for mobile crisis units.

Variable	Type	Example Value	Description
timestamp	datetime	2025-12-17 02:57:00	Initial date and time the crisis call was logged.
incident_id	double	13209	Unique identifier for the crisis incident.
agency	factor	MCS LC	The specific Mobile Crisis Service agency responding to the call.
city	factor	Springfield	The city or municipality where the incident occurred.
call_type	factor	Agitation	The primary clinical nature or behavioral reason for the call.
call_outcome	factor	Remained in community	The final disposition or result of the intervention.
dispatch_time	datetime	2025-12-17 03:14:00	Time when the mobile crisis team was officially dispatched.
client_engagement_time	datetime	2025-12-17 03:38:00	Time when the team made direct clinical contact with the client.
arrival_time	datetime	2025-12-17 03:35:00	Time when the mobile crisis unit arrived at the scene.
clear_time	datetime	2025-12-17 03:48:00	Time when the unit completed the intervention and became available again.
dispatch_status	factor	Engaged	Status of the dispatch (e.g., Engaged, Refused).
minutes_request_dispatch	double	17	Duration in minutes between the initial request and dispatch.
minutes_dispatch_arrival	double	21	Duration in minutes between dispatch and arrival on scene.
minutes_arrival_engagement	double	3	Duration in minutes between arrival and start of client engagement.
minutes_arrival_departure	double	13	Total time spent on scene in minutes (arrival to departure).

4 Overview and exploratory analysis

The following analysis will focus on investigating the changes in Eugene's response to non-violent emergencies related to homelessness, addiction and mental health after MCS LC's launch on August 18th 2024 and CAHOOTS' discontinuation on April 7th 2025 in Eugene.

4.1 Summary

Let's begin with a generic overview and exploratory analysis of the data.

```
## Rows: 1,601,142
## Columns: 12
```

```
## $ timestamp       <dttm> 2015-01-01 00:00:00, 2015-01-01 00:00:44, 2015-01-01 ~
## $ incident_id     <chr> "cad_15000001", "cad_15000002", "cad_15000003", "cad_1~
## $ EPD             <fct> 1, 1, 0, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 0, ~
## $ SPD             <fct> 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, ~
## $ CAHOOTS         <fct> 0, 0, 1, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 1, ~
## $ MCSLC           <fct> 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, ~
## $ city            <fct> Eugene, Eugene, Eugene, Eugene, Eugene, Eugene, Eugene~
## $ nature          <fct> Person Stop, Fight, Check Welfare, Shots Fired, Illega~
## $ outcome         <fct> Assisted, Resolved, Assisted, Patrol Check, Advised, P~
## $ dispatch_status <fct> Non MCSLC, Non MCSLC, Non MCSLC, Non MCSLC, Non MCSLC,~
## $ priority        <fct> 6, P, 5, 3, 5, 6, P, 3, 6, 4, 3, P, 6, 6, 6, 3, 3, 1, ~
## $ source          <fct> CAD, CAD, CAD, CAD, CAD, CAD, CAD, CAD, CAD, CAD, CAD,~
```

4.2 Global Call Volume

Table 1: Global repartition of call volume per city and agency

city	Total_calls	EPD	SPD	CAHOOTS	MCSLC
Eugene	1031525	905197	0	125652	7097
Other	1100	0	0	0	1100
Out of County	3	0	0	0	3
Springfield	568239	0	527491	52508	710
Unknown	275	0	0	0	275

4.3 Frequent Natures

The most frequent natures for CAHOOTS calls are Public Assist and Check Welfare. These will be our main focus for the whole analysis, along with some other call natures appearing in the plot. Total, we will use the top 14 most frequent natures in calls handled by CAHOOTS in both Eugene and Springfield, and the mcslc natures (14).

4.4 Evolution In Time

Here is a timeline of CAHOOTS' and MCSLC's dispatched calls evolution between 2015 and 2025. We can see that right after CAHOOTS stopped, MCSLC's number of dispatches went up.

4.5 Broad Categories Overview

Next is a general overview of broad outcome categories for every existing call nature, for the whole dataset (EPD, SPD, MCSLC and CAHOOTS responses).

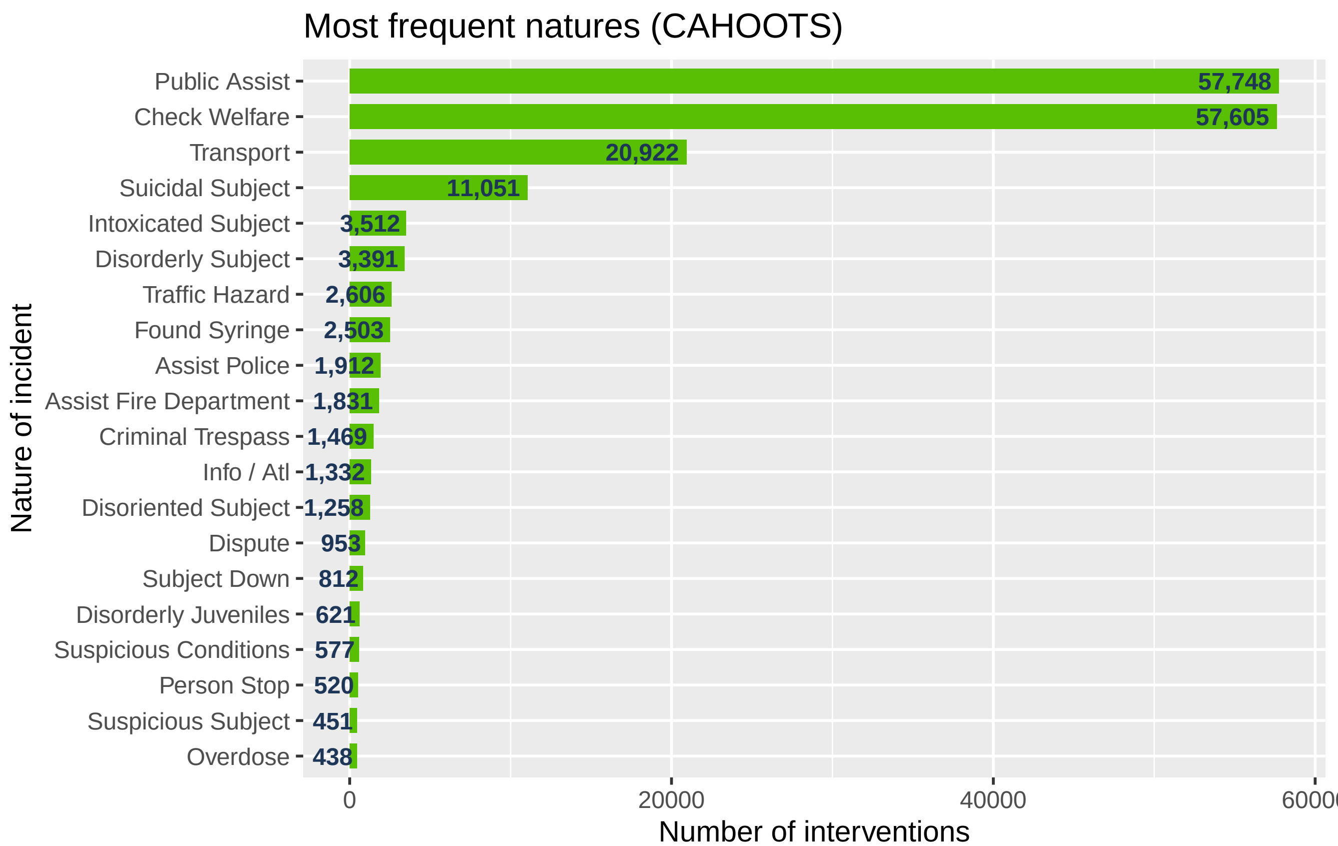


Figure 1: Top 10 most frequent call natures for CAHOOTS dispatches

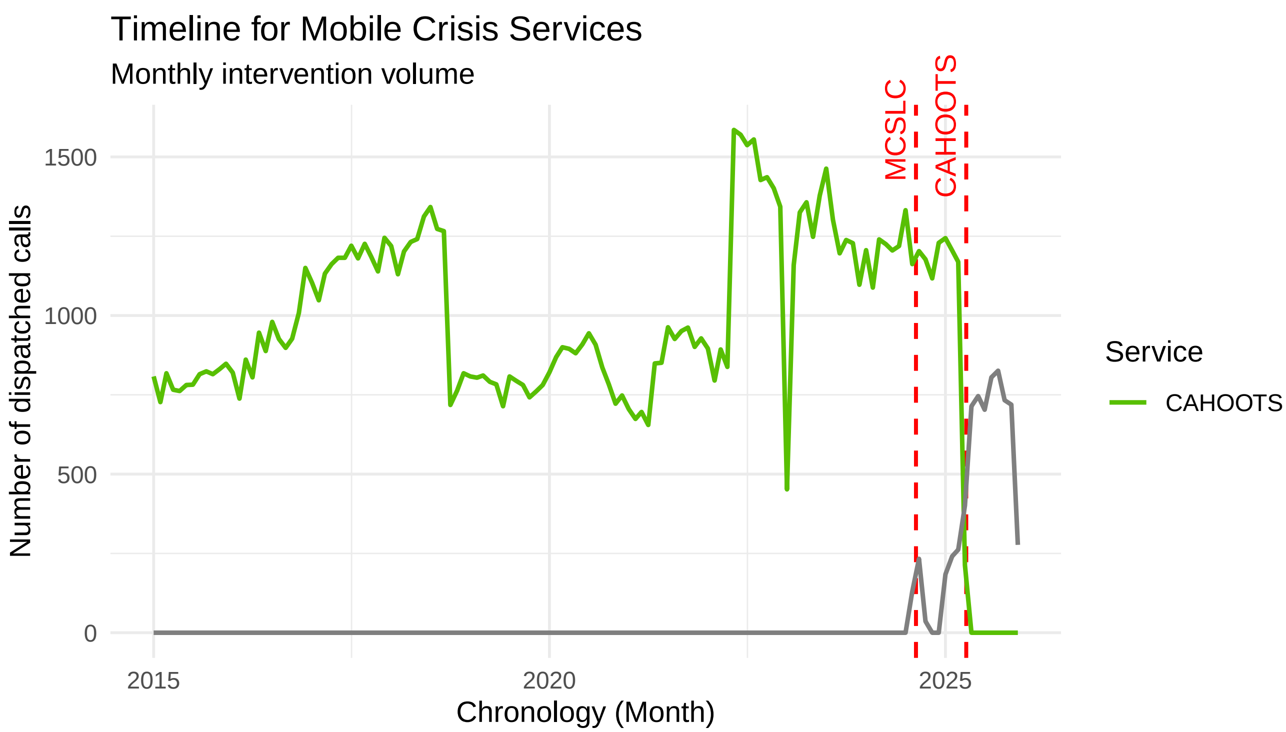
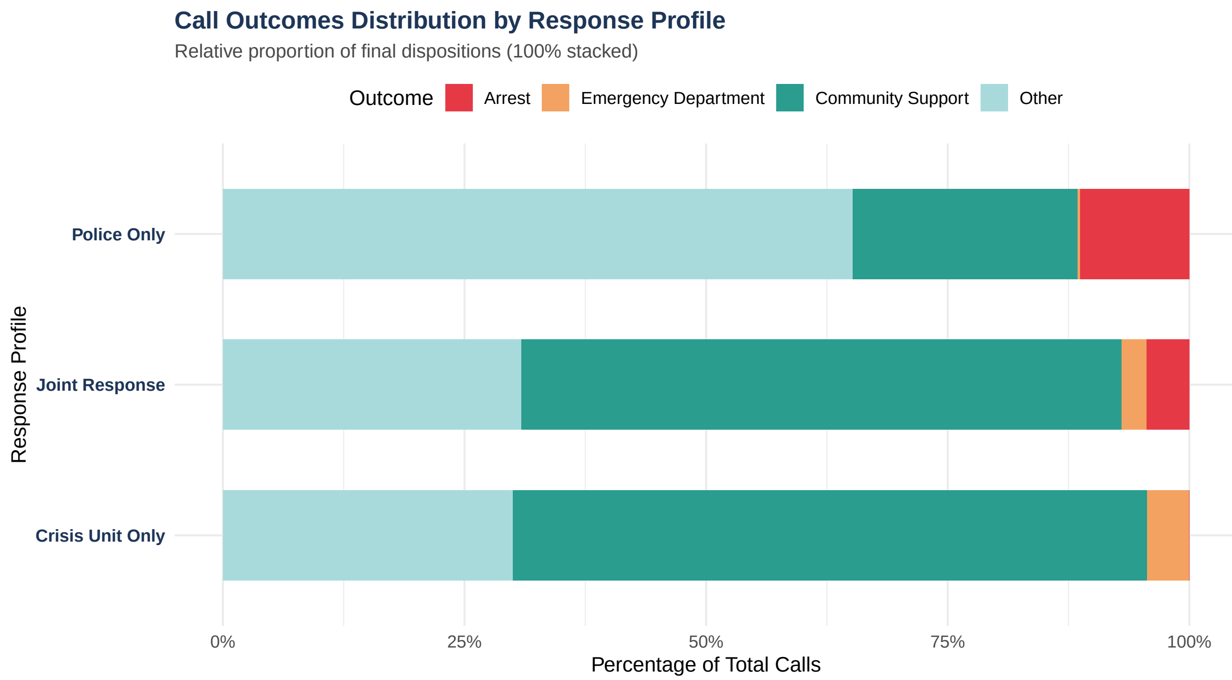


Figure 2: Monthly evolution of call volume by mcs agency (2015-2025)



4.6 Exploratory Analysis of Call Outcome

To better understand the effects of CAHOOTS' discontinuation and MCSLC's roll-out, we will be focusing on the time period where they were both active, and then the period during which CAHOOTS stopped servicing Eugene. This will allow us to see if there is any change in MCSLC's activity, if other agencies, like the EPD, took over some calls that CAHOOTS used to handle, and how it impacted call outcomes in general.

4.6.1 Temporal Filtering and Harmonization

First, let's filter the data to keep only the relevant time periods: - CAHOOTS and MCSLC active: August 18, 2024 to April 7, 2025 - After CAHOOTS stopped services: April 8, 2025 to present

To do so, we define the temporal boundaries in datetime variables. We then group the data into weekly bins to smooth out daily fluctuations, and we sort the the calls into the `Overlap` and the `Post-CAHOOTS` categories, respectively the MCSLC and CAHOOTS joint period, and the only MCSLC period. We exclude all historical data, before August 18, 2024 when MCSLC started to maintain a strictly controlled analytical scope.

The overlap period is 232 days long, and the post-CAHOOTS period has 254 days.

We have 9038 dispatches for CAHOOTS and 1116 dispatches for MCSLC during the overlap period. During the post-CAHOOTS period, we have 5894 MCSLC dispatches.

Here is an overview of the filtered data.

For MCLSC dispatches, we have the 7 following possible outcomes. Most of dispatched calls end with **Remaine In Community**, and we note that some calls end with **Arrest**.

outcome	n
Remained In Community	7083
Emergency Department	1559

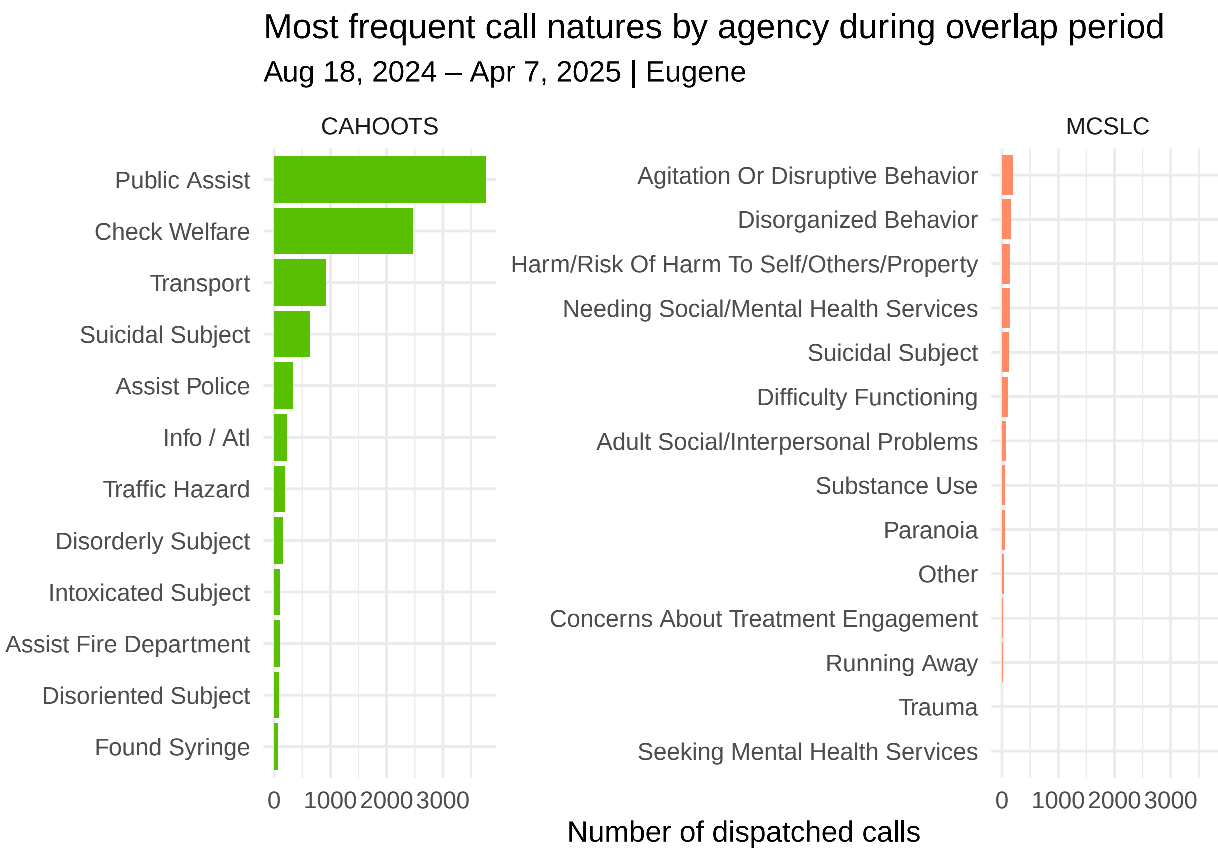
outcome	n
Other	323
Sobering Or Detox Facility	62
Arrest	38
Respite	7
Crisis Walk-In Center	4

CAHOOTS only calls have 35 possible outcomes. Here are the 10 most frequent ones, with **Assisted** and **Unable To Locate being the most frequent ones**. For CAHOOTS, most of the interventions are closed as **Assisted** with no further detail. Some outcomes are from CAHOOTS and EPD joint responses. EPD interventions are often closed with different outcomes for the same type of call, we can assume that these outcomes are similar but just registered under a different name.

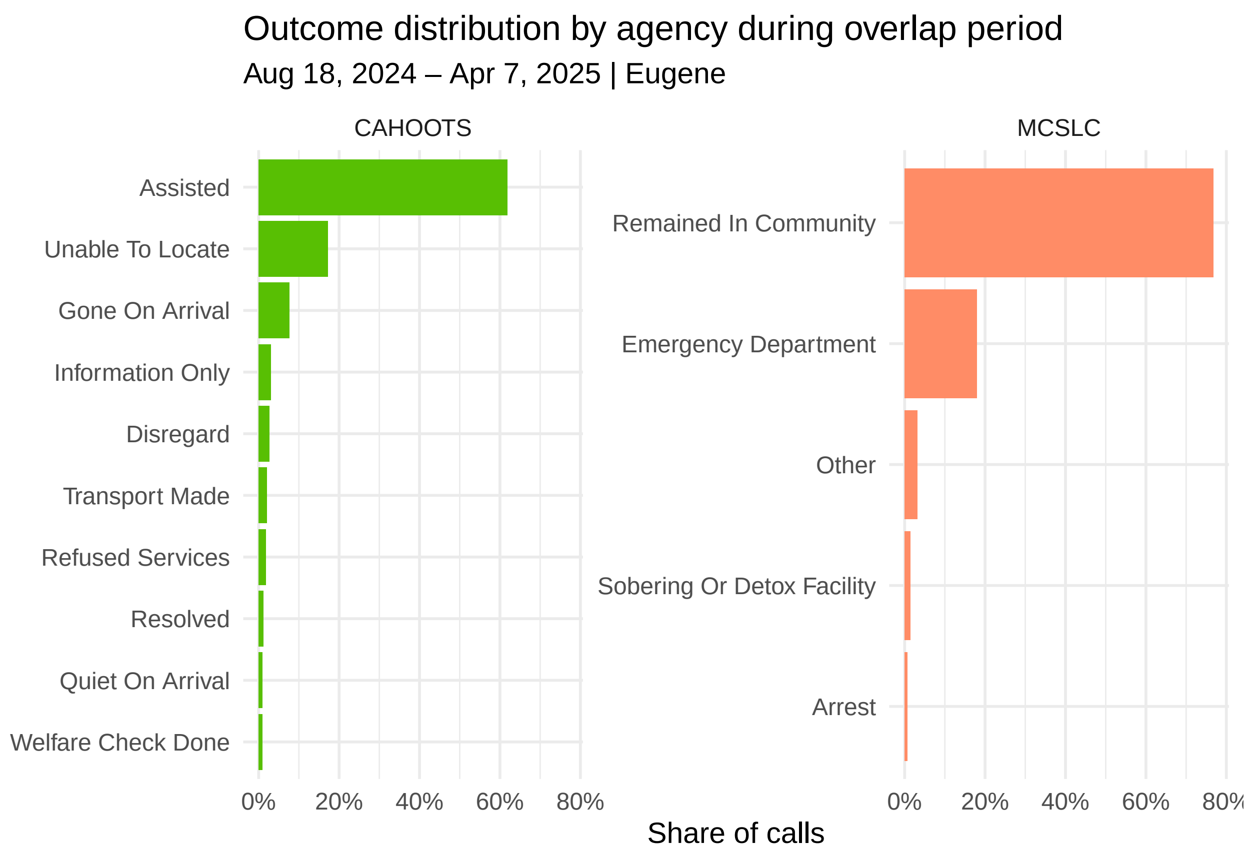
outcome	n
Assisted	7721
Unable To Locate	2063
Gone On Arrival	824
Disregard	324
Information Only	303
Transport Made	239
Refused Services	178
Welfare Check Done	176
Resolved	171
Refused Services (Cahoots)	157
Advised	134
Report Taken	116
Quiet On Arrival	97
Arrest	24
Referred To Other Agency	15
Patrol Check	12
Duplicate Event	11
Unfounded	10
Non Criminal Hold	9
Cited In Lieu Of Custody	7
Warning	6
No Investigation	5
Disregarded By Dispatch	4
Follow Up Investigation	4
Caller Will Call Back	3
Canceled Report Number	3
Caller Called Back	2
Returned To Owner	2
Uniform Traffic Citation Issued	2
Civil Issue	1
Disregarded By Patrol Supervisor	1
False Alarm	1
No Patient Transport	1
Parking Citation Issued	1
Sobriety Check	1

4.6.2 CAHOOTS vs MCSLC

4.6.2.1 Overlap Period Comparison The goal here is to look at CAHOOTS and MCSLC's types of services. We extract the most frequent call natures for both agencies. For this time period and city, CAHOOTS has 12 call natures, with **Public Assist** and **Check Welfare** being the most common ones. MCSLC has 14, **Agitation or Disruptive Behavior** and **Disorganized Behavior** are the two most common.



Here is the call outcome distribution by agency during the overlap period.



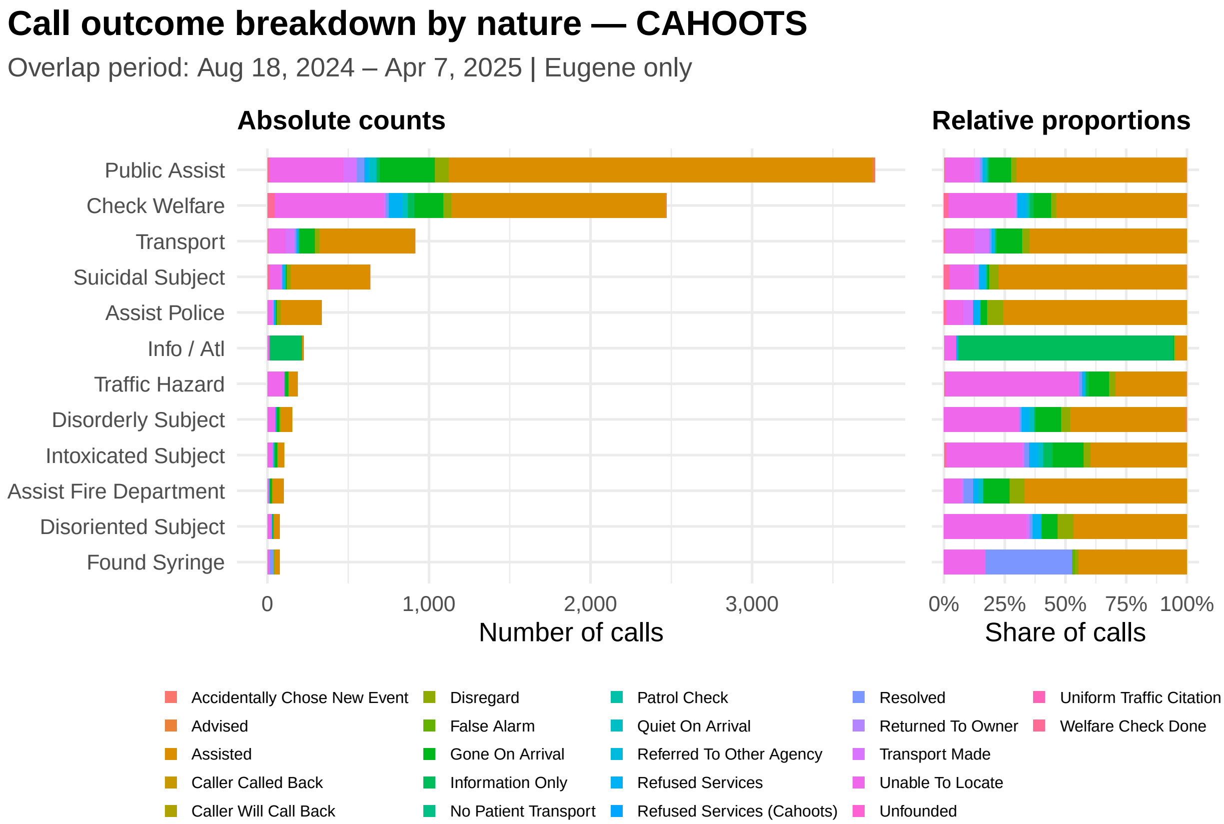
If we reuse the broad outcome category from the exploratory analysis, we find the following numbers. We will dive in deeper and more precisely in the following subsections of this analysis.

Table 2: Broad outcome category distribution by agency during overlap period (Eugene, Aug 2024 – Apr 2025)

Agency	Outcome	Count	Share
CAHOOTS	Community Support	5817	64.4%
CAHOOTS	No Contact	2491	27.6%
CAHOOTS	Other	560	6.2%
CAHOOTS	Refused Services	168	1.9%
CAHOOTS	Arrest	2	0.0%
MCSLC	Community Support	857	76.8%
MCSLC	ED / Crisis Facility	216	19.4%
MCSLC	Other	36	3.2%
MCSLC	Arrest	7	0.6%

4.6.2.2 Breakdown of Call Outcome During the overlap period, CAHOOTS handled a majority of **Public Assist** and **Check Welfare** calls, representing over half of all CAHOOTS calls in Eugene for this time period. The dominant outcome across almost all natures is **Assisted**, confirming that most CAHOOTS interventions resulted in direct on-scene support without escalation.

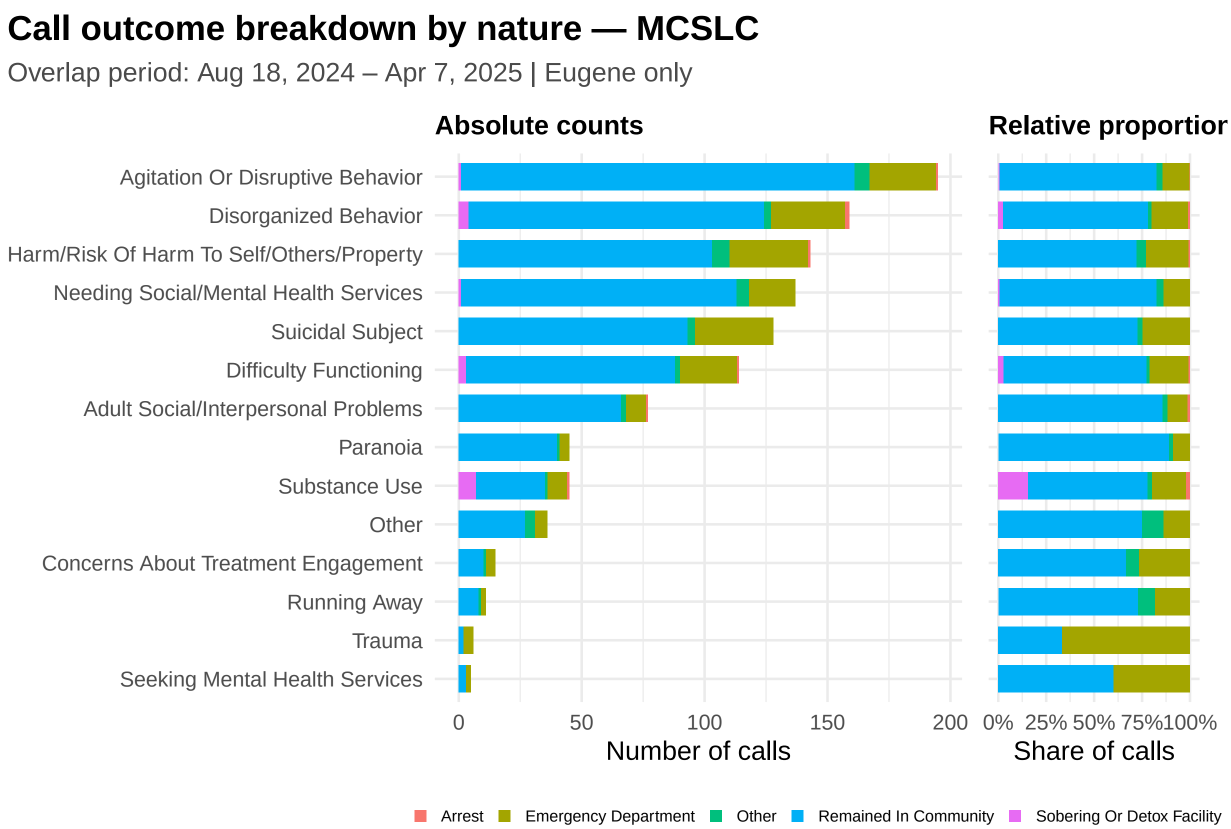
However, notable exceptions appear for **Info/Atl** and **Disoriented Subject**, where outcomes like **Gone On Arrival** and **Unable To Locate** are more frequent. **Arrest** is rare across all natures.



MCSLC's call volume during the overlap period is lower than CAHOOTS'. It could be due to its more recent launch and its separate dispatch pathway (they are outside of the 911 CAD system). While MCSLC's goal is to manage mental health, behavioral, and substance abuse crisis as well, their scope of action stays different than CAHOOTS'.

The dominant outcome across all natures is **Remained in Community**. **Emergency Department** transport appears as a secondary outcome, especially for higher-risk natures such as **Suicidal Subect** and **Harm/Risk of Harm** calls.

We also see some **Arrest** outcomes for some call natures, indicating a joint response from MCSLC and the EPD, even though they do not appear as responding agency in the data. MCSLC being on its own call line is not jointly dispatched along with the police, but they coordinate with other agencies if needed (EPD, other existing care providers…).



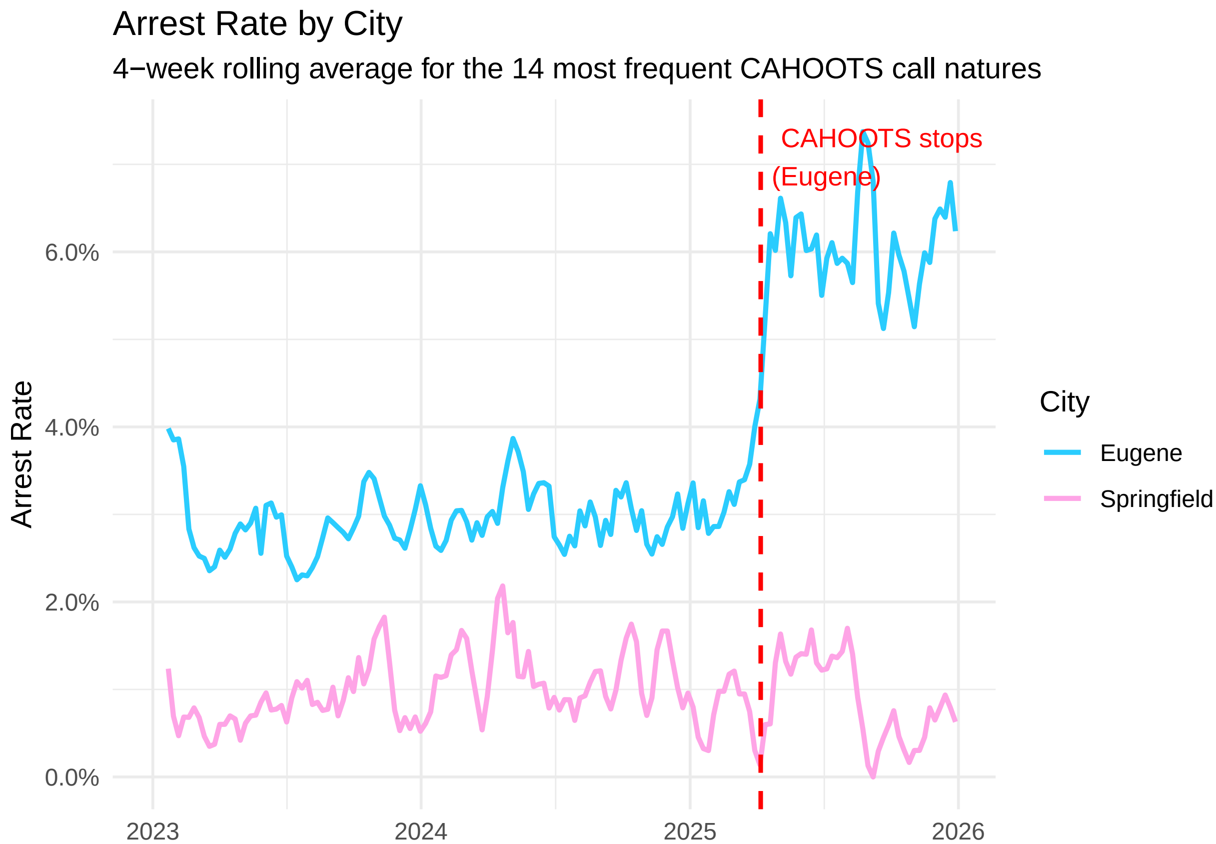
5 Post-CAHOOTS Call Outcome Evolution in Eugene

5.1 Arrests

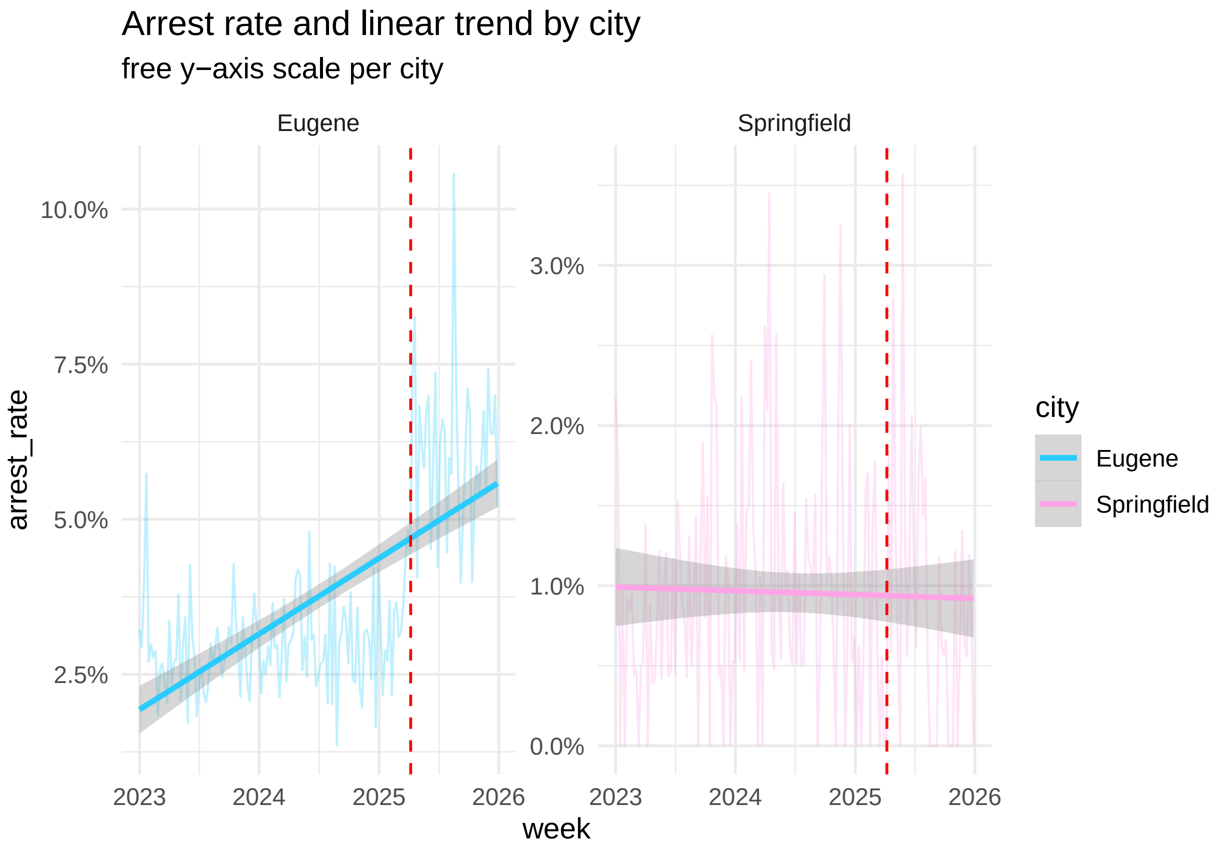
Now moving on the the call outcome evolution analysis, we start by harmonizing the data. We calculate the weekly rolling average arrest rate per week for Eugene and Springfield, using only CAHOOTS call types regardless of the agency in charge of the intervention.

5.1.1 Parallel Trends

5.1.1.1 Line Plot Here is a plot of the trends side by side for a visual overview. We can see that the arrest rates trend for each city is quite similar before the discontinuation, but Eugene's volume is superior. After CAHOOTS stopped servicing Eugene, we observe a surge in arrest whereas Springfield's stays similar.



5.1.1.2 Linear Trends The linear trends shows with more precision that Eugene saw an increase in arrests and Springfield's rate is relatively constant over time.



5.1.1.3 Statistical Test We then test the parallel trends more rigorously to see if we can use the Differences in Differences method.

H0 : both groups are the same before treatment (parallel trends) H1 : trends are different

we don't want to reject null hypothesis so p-value should be over 0.5 (for 95% certainty)

Here p-value is 0.544 so the parallel trends passed, meaning that before CAHOOTS was discontinued, arrest rate trends were similar in Eugene and Springfield for call natures typically handled by CAHOOTS, regardless of the agency in charge. We can proceed with the DiD method.

```
##
## Parallel Trends Check
## ===============================================
##                         Dependent variable:
##                     ---------------------------
##                             Arrest Rate
## -----------------------------------------------
## Week Number                   0.00002
##                              (0.00003)
##
## Treated                      0.019***
##                               (0.002)
##
## Week Number:Treated           0.00002
##                              (0.00003)
```

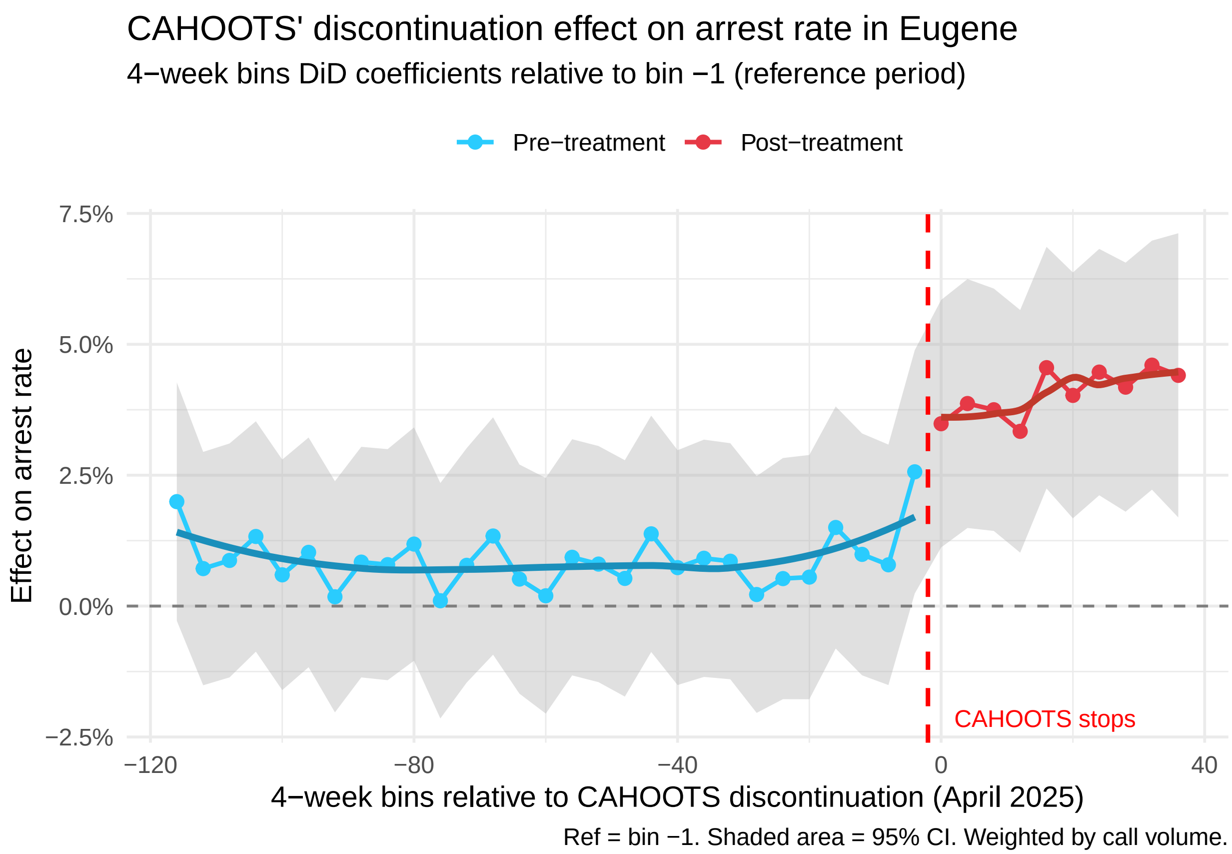
```
##
## Constant                      0.009***
##                                (0.002)
##
## -----------------------------------------------
## Observations                    236
## R2                             0.611
## Adjusted R2                    0.606
## Residual Std. Error      0.139 (df = 232)
## F Statistic          121.234*** (df = 3; 232)
## ===============================================
## Note:               *p<0.1; **p<0.05; ***p<0.01
```

5.1.2 Differences in Differences

We test the changes in arrest rate using the DiD method. We find (95% confidence level) an increase of 3.2 percentage points for weekly arrest rate in Eugene compared to Springfield for same period after treatment (CAHOOTS stop)

```
##
## DiD, Effect of CAHOOTS Discontinuation on Arrest Rate (Weekly)
## ==================================================================
##                                         Dependent variable:
##                                    -------------------------------
##                                              Arrest Rate
## ------------------------------------------------------------------
## Eugene (Treated)                              0.020***
##                                                (0.001)
##
## Post-Discontinuation                          -0.0004
##                                                (0.002)
##
## DiD Estimator (Treated × Post)                0.032***
##                                                (0.003)
##
## Constant                                      0.010***
##                                                (0.001)
##
## ------------------------------------------------------------------
## Observations                                     314
## R2                                              0.774
## Adjusted R2                                     0.771
## Residual Std. Error                       0.154 (df = 310)
## F Statistic                           353.036*** (df = 3; 310)
## ==================================================================
## Note:                                  *p<0.1; **p<0.05; ***p<0.01
##                                    OLS weighted by weekly call volume.
```

5.1.2.1 Event Study If we take a closer look at the coefficients, grouped by bins of 4 weeks, we can calculate the confidence intervals and plot the arrest rate rolling average, the DiD coefficients.



5.1.2.2 Robustness Test Arrests Try different time window to see if it impacts previous results (it passed)

```
##
## ==========================================================================================
##                                                    Dependent variable:
##                                     ------------------------------------------------------------
##                                                        Arrest Rate
##                                        Full period (2023-2025)        Narrow window (±12 months)
##                                                 (1)                            (2)
## ------------------------------------------------------------------------------------------
## Eugene (Treated)                            0.020***                       0.026***
##                                             (0.001)                        (0.005)
##
## Post-Discontinuation                        -0.0004                         0.006
##                                             (0.002)                        (0.006)
##
## DiD Estimator (Treated × Post)              0.032***                       0.021***
##                                             (0.003)                        (0.007)
##
## Constant                                    0.010***                        0.008*
##                                             (0.001)                        (0.004)
##
## ------------------------------------------------------------------------------------------
## Observations                                  314                             50
```

```
## R2                                              0.774                           0.785
## Adjusted R2                                     0.771                           0.771
## Residual Std. Error                        0.154 (df = 310)                0.180 (df = 46)
## F Statistic                            353.036*** (df = 3; 310)        55.840*** (df = 3; 46)
## ==========================================================================================
## Note:                                                              *p<0.1; **p<0.05; ***p<0.01
##                                 Robustness check: restricting to ±12 months around discontinuation.
```

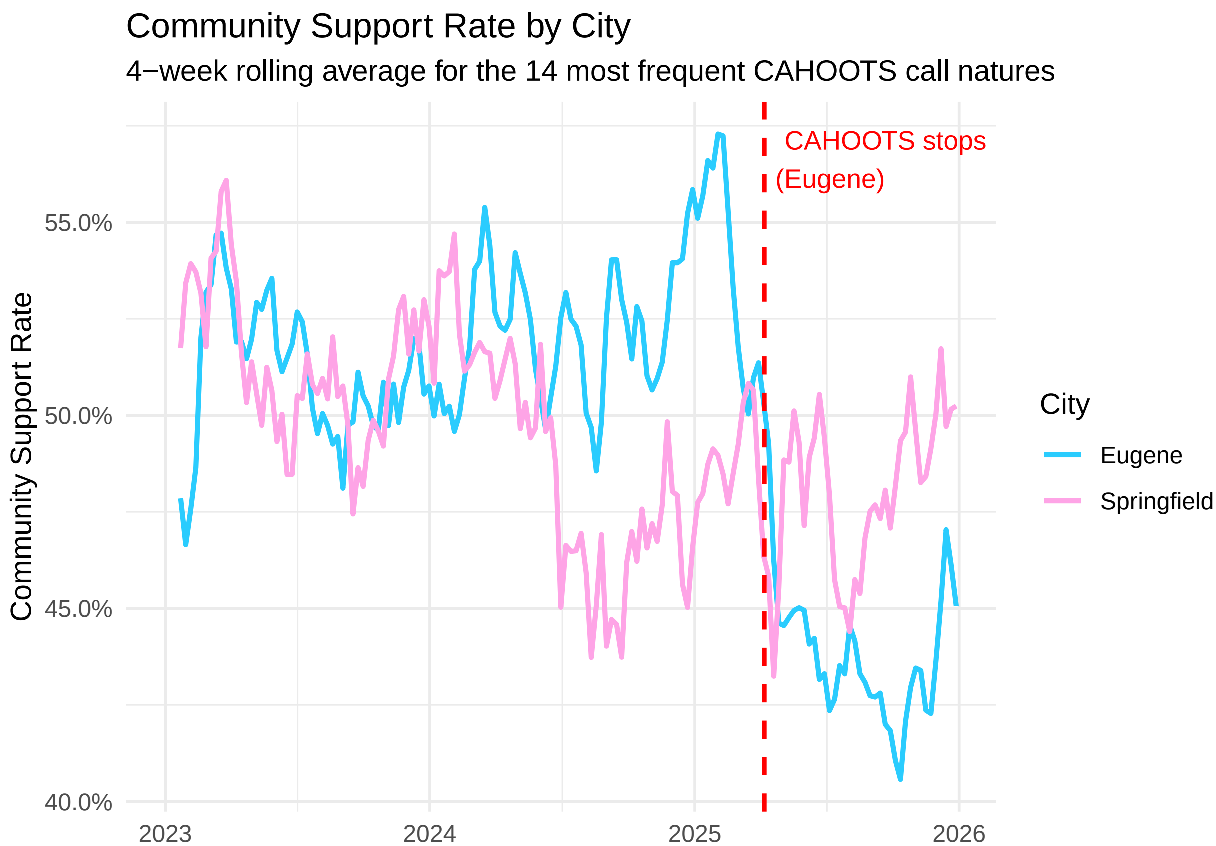
TO DO THURSDAY

- put graph in report
- explain method in report
- add theory behind method and statistical testing (equations etc)
- do DiD but for other outcomes
- write State of the Art
- wrap up

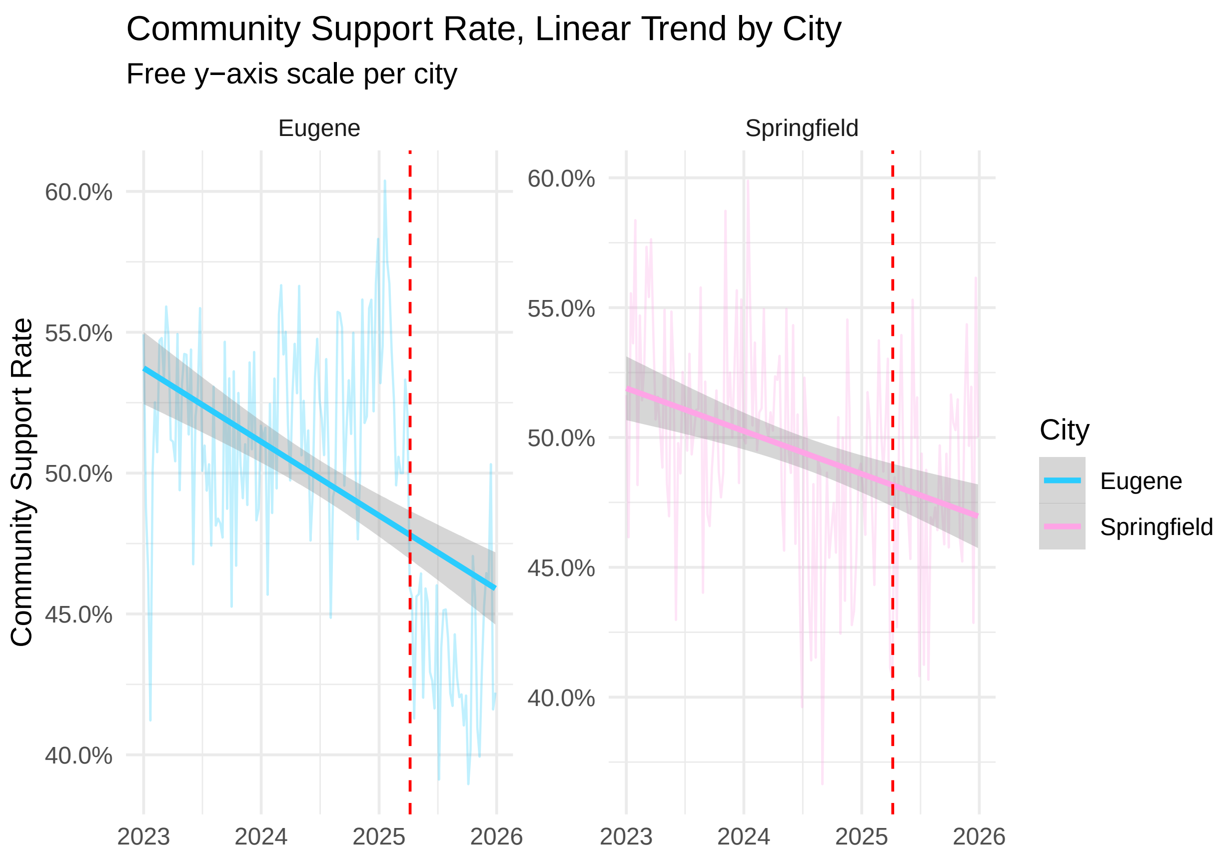
5.2 Community Support

Let's take a look at welfare checks and similar calls. The outcome associated with welfare checks are `Assisted` for CAHOOTS and `Welfare Check Done` for the police. The `Assisted` call outcome is also used by CAHOOTS for other calls when the outcome was successful.

5.2.1 Line Plot



5.2.2 Linear trends



5.2.3 Parallel trends check

*** passed

```
##
## ===============================================
##                          Dependent variable:
##                      ---------------------------
##                            community_rate
## -----------------------------------------------
## week_num                     -0.0005***
##                               (0.0001)
##
## treated                       -0.019**
##                                (0.009)
##
## week_num:treated               0.001***
##                               (0.0001)
##
## Constant                       0.527***
##                                (0.008)
##
## -----------------------------------------------
```

```
## Observations                        236
## R2                                 0.137
## Adjusted R2                        0.126
## Residual Std. Error      0.632 (df = 232)
## F Statistic           12.317*** (df = 3; 232)
## ==================================================
## Note:               *p<0.1; **p<0.05; ***p<0.01
```

5.2.4 DiD Community Support

- 6.3 pp community support outcome in dispatched calls for Eugene after CAHOOTS stopped (*** significant)

```
##
## DiD, Effect of CAHOOTS Discontinuation on Community Support Rate (Weekly)
## ===================================================================
##                                           Dependent variable:
##                                   -----------------------------------
##                                         Community Support Rate
## -------------------------------------------------------------------
## Eugene (Treated)                              0.018***
##                                                (0.005)
##
## Post-Discontinuation                          -0.020**
##                                                (0.009)
##
## DiD Estimator (Treated × Post)                -0.063***
##                                                (0.010)
##
## Constant                                      0.500***
##                                                (0.004)
##
## -------------------------------------------------------------------
## Observations                                     314
## R2                                              0.406
## Adjusted R2                                     0.400
## Residual Std. Error                       0.621 (df = 310)
## F Statistic                         70.600*** (df = 3; 310)
## ===================================================================
## Note:                                   *p<0.1; **p<0.05; ***p<0.01
##                                   OLS weighted by weekly call volume.
```

6 Appendix

6.1 Abbreviations

- SPD = Springfield Police Department
- CAD = Computer-Aided Dispatch
- MCS LC = Mobile Crisis Services Lane County
- EPD: Eugene Police Department

- CAHOOTS: Crisis Assistance Helping Out On the Streets
- MCSLC : Mobile Crisis Services Lane County
- SPD: Springfield Police Department
- MCR: Mobile Crisis Response